MIL-A-7237B(AV)
4 February 1975
Superseding
MIL-A-7237A
31 July 1962

MILITARY SPECIFICATION

AIR INDUCTION SYSTEM, RECIPROCATING ENGINE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an air induction system for reciprocating engine-powered aircraft.

1.2 Classification. The reciprocating engine air induction system shall be of the following types, as specified (see 6.3):

Type I - Nonsupercharged.
Type II - Supercharged.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for approval, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-F-7194	Filters, Reciprocating Engine Induction Air, Aircraft

FSC 2945

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

MIL-I-8599	Installation of Reciprocating Engine Induction Air Filters, Design Requirements for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-E-25109	Engines, Aircraft, Reciprocating, General Specification for
MIL-D-1000	Drawings, Engineering and Associated List

STANDARDS

FEDERAL

FED-STD-151	Metals: Test Methods

MILITARY

MIL-STD-130	Identification Markings of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS28034	Bulbs, Temperature, Electrical Resistance (Minus 70 Deg. C. to Plus 300 Deg. C.).

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

NATIONAL AERONAUTICAL AND SPACE ADMINISTRATION

Report No. 1235 — Standard Atmosphere - Tables and Data for Altitudes to 65,800 Feet

(Application for copies should be addressed to the National Technical Information Service, Springfield, VA 22151).

3. REQUIREMENTS

3.1 Design of air induction system. Design of air induction system for reciprocating engine-powered aircraft shall be of the type (see 2.1) as specified in 6.3.

3.1.1 Type I. Nonsupercharged air induction system shall consist of engine induction air filter, ducting, induction air heat valves for control of airflow, flexible joints, and controls.

3.1.2 Type II. Supercharged air induction system shall consist of engine induction air filter, ducting, drain valve, induction air heat valves for control of airflow, supercharger intercooler, flexible joints, temperature bulb, fuel metering unit deck screens, and controls.

3.2 Preproduction sample. A sample air induction system installation shall be subjected to the preproduction tests specified in 4.4. Approval of the preproduction installation design by the activity concerned shall not relieve the contractor of his responsibility to supply air induction system installations conforming to this specification. Any changes or deviations from the approved preproduction installation shall be subjected to the approval of the contracting officer.

3.3 Materials. Materials and processes used shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used, provided the specifications are approved by the procuring activity and contain provision for adequate tests. The use of contractor's specifications shall not constitute waiver of Government inspection.

3.3.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service. The use of any protective coating that will crack or scale with age or extremes of climatic and environmental conditions shall be avoided. Metals shall be inspected in accordance with FED-STD-151.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

3.3.1.1 Dissimilar Metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.2 Fungusproof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. If any materials are nutrients for fungi, they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.4 Components. Principal components of the air induction system follow. (Typical air induction systems and arrangement of components are shown on figs. 1, 2, and 3.)

3.4.1 Air intake system. Engine air induction system shall provide a means of obtaining air from outside the aircraft and conducting it to the fuel metering unit or throttle body of the engine, for the selection of air from a combination of the following sources: ram or nonram, heated or nonheated, filtered or nonfiltered. The selection of air to the engine shall include, but not necessarily be limited to, those sources. Filtered air shall be obtained by passing the intake air through an air cleaner to remove foreign particles. Filtered intake air shall be heated by passing through a preheating chamber to raise the temperature of the air to eliminate ice formation in critical areas of the air induction system and to aid engine operation at low temperatures. The engine air, obtained from outside the engine cowling, shall be taken from point(s) of the engine in a low dust concentration area. Flexible metal tubing or backfire doors shall not be incorporated in the system.

3.4.1.1 Duct. The air intake duct shall not be subject to failure as a result of backfire, vibration, or flexing of the aircraft structure and engine mount. The duct shall be mounted to the airframe and connected to the engine with flexible connections.

3.4.1.2 Ram air intakes. Ram air intakes shall be located in a region of high impact pressure. The intakes shall consist of a ram air scoop and ducting as necessary to conduct the air to the carburetor fuel metering unit of the engine with a minimum of pressure loss.

3.4.1.2.1 Type I - nonsupercharged. Ram air intakes shall be such that the total pressure head available at the deck of the throttle body, in level and climbing attitudes at military critical altitudes, is not less than 75 percent of the aircraft free stream total pressure head.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

3.4.1.2.2 Type II - supercharged. The supercharged system shall maintain a predetermined air intake pressure to the fuel metering unit at any altitude, within the limits of the type supercharger incorporated in the system. Superchargers shall be in accordance with Specification MIL-E-25109. The total pressure head at the supercharger inlet shall not be less than 75 percent of the aircraft free stream total pressure head, in level or climbing attitude at military power and critical altitude, when tested in accordance with 4.4.

3.4.1.2.2.1 Intercooler. If required in the type II system, an intercooler shall be provided between the supercharger and the fuel metering unit to lower the air temperature from the supercharger to the engine.

3.4.1.2.2.1.1 Efficiency. The capacity and the installation of the intercooler shall provide for an efficiency of not less than 60 percent when tested in accordance with 4.4.9. The procuring activity shall be furnished with effectiveness data for the intercooler, as shown on figure 4.

3.4.1.2.2.1.2 Installation. The intercooler shall be mounted so that vibration failure of the intercooler, or its mounting supports, does not occur when tested in accordance with 4.4.10. Expansion or contraction of the intercooler shall not be unduly restricted. Stops shall be incorporated on supporting cradles used to mount the intercooler, to prevent damage due to shock loading.

3.4.1.2.2.1.3 Construction. Guard strips shall be used to protect intercooler where the sides of wall core tubes or edges of fins form the face of the intercooler, and other easily damaged parts. These strips shall support the intercooler and shall not be spaced more than 4 inches apart.

3.4.1.2.2.1.4 Airflow. An intercooler shutter shall be provided at the cooling air exit of the core, where possible. The shutter shall be free of resonant vibration in the range of all aerodynamic exciting force. The shutter shall not cause aerodynamic disturbance in any flight attitude or power conditions which affect performance.

3.4.1.2.2.1.5 Pressure. The intercooler and its connections shall not deform more than 0.08-inch when an intermittent pressure of 25 pounds per square inch (psi) is applied.

3.4.2 Air filters. Engine induction air filters shall be provided as permanent equipment in all air induction systems. The filters shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

constructed in accordance with Specification MIL-F-7194 and installed in accordance with Specification MIL-I-8599.

3.4.3 Heat system. The air induction heat system shall consist of all the parts necessary for a complete installation on the particular aircraft designated (see 6.3(c)).

3.4.3.1 Control. A single control shall be provided to select air to the fuel metering unit and to provide for the metering of heated air increments not greater than 20°F. The "full cold" position on the cockpit control shall assure that no hot air enters the fuel metering unit. The cockpit control shall be operative in either the "full hot" or "full cold" position from any intermediate position on the control. The maximum force required to actuate the air induction system hand control knob for each engine shall not exceed 10 pounds under any operating condition. This force shall be the total of the force necessary to overcome friction in the entire control systems. The force necessary to actuate the induction components is measured at the center of the hand control knob with an incorporated locking device in the unlocked loose position. The maximum force specified shall be applicable in either direction of movement and under all conditions of operation from plus 160° to minus 65°F.

3.4.3.2 Temperature rise. Installation supplying heated air to the carburetor(s) shall be capable of providing inlet air at a minimum temperature of 70°F. to the carburetor, within 30 seconds at sea level, when the outside air temperature is minus 40°F. and the engine is developing 65 percent normal sea level rated power. Installations having a fuel metering unit other than a carburetor shall maintain a fuel metering air inlet temperature of plus 40°F. when the outside air temperature is minus 40°F. and the engine is developing 65 percent of its normal rated sea level power. The induction air heat system shall maintain a minimum fuel metering air inlet temperature of 0°F. when the outside air temperature is minus 65°F. and the engine is operating at idle revolutions per minute (rpm) at sea level conditions.

3.4.3.3 Temperature variation. The maximum allowable variation of the average air temperature between the fuel metering inlet, automatic temperature compensating unit, and the temperature indicating unit shall not exceed 5°F.

3.4.3.4 Pressure variation. The static pressure at the inlet of the fuel metering unit shall not be reduced more than 2 inches of mercury when the induction air heat control is switched from "full cold" position to "full hot" position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

3.4.4 Fuel metering unit screen. An air screen of 0.250-inch mesh shall be provided for systems having engines with integral mechanical superchargers (engine-driven blowers). The screen shall be installed with the unit, at the entrance to the fuel metering unit. The junction at this point shall be fueltight. The screen shall be readily accessible for cleaning.

3.4.5 Temperature bulb. A temperature bulb conforming to MS28034 shall be provided to measure air temperature at the entrance to the fuel metering unit. The standard indicator shall be visible to the pilot, co-pilot, and flight engineer. The bulb shall be installed at least 1.5 inches upstream from the entrance to the fuel metering unit, excluding the automatic mixture compensating device, and the instrument shall indicate variations in temperature for any change in setting of the heat control throughout the full range from hot to cold. Care shall be taken to insure that the hot air inlets and valves do not affect the bulb readings. The bulb installation shall be checked against a thermocouple installed on the carburetor deck screen and the difference in readings shall be recorded and submitted to the procuring activity. Installation for this calibration shall be in accordance with figure 5.

3.4.6 Auxiliary air bleeds. Auxiliary air bleeds may be provided by obtaining prior approval from the procuring activity.

3.4.7 Valves. Valves used for the control of airflow shall be installed in a manner that prevents water or fuel from collecting and entering the induction system. The valves shall not become inoperative by icing.

3.5 Construction.

3.5.1 Dimensions and weight. Dimensions of the air induction system shall be governed by the type engine installed in the specific aircraft designated in 6.3. Total weight of the system, including components, shall be as light as practicable.

3.5.2 Standards and specifications. Standards and specifications for commodities and services not specified herein, shall be selected in accordance with MIL-STD-143.

3.5.2.1 Standard parts. AN and MS standard parts shall be used where they suit the purpose. Commercial parts, such as fasteners, connectors, and cotter keys may be used, provided they can be replaced by AN and MS

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

standard parts without alteration or degradation of function. The parts list shall reference the applicable AN and MS standard parts as acceptable replacements for the commercial parts.

3.5.3 Interchangeability of parts. Components shall be interchangeable (without requiring modification for replacement) with similar components furnished under the same contract or order.

3.6 Performance. The air induction system shall satisfy the performance requirements of 4.3 and 4.4.

3.6.1 Leakage. The complete induction system leakage shall not exceed 0.5 percent of the maximum engine airflow at military rated power.

3.6.2 Pressure. The complete induction system shall withstand an internal air pressure of 25 pounds per square inch gage (psig) without rupture, distortion, or other defects affecting performance or service life of the system.

3.6.3 Distribution. The air inlet system shall maintain a uniform velocity distribution of air to the fuel metering unit inlet under all flight conditions.

3.6.4 Uncompensated venturi suction. The uncompensated venturi suction for any value of the airflow and air density encountered in flight shall not vary more than $\pm$ 2 percent from the corresponding fuel metering unit value defined by the applicable engine model specification at the time the setting was established when using the normal air supply (ram air). The uncompensated venturi suction for any airflow and air density encountered in flight when using an alternate air system (filtered air or heated air) shall not vary more than $\pm$ 2 percent of the value obtained when using normal air supply system.

3.7 Drawings. Complete class II detailed engineering drawings of the air induction system, including all dimensions, shall be submitted in accordance with Specification MIL-D-1000, to the procuring activity prior to date for testing. One set of reproducible (Van Dyke or equivalent) detail drawings shall also be furnished.

3.8 Finish. All aluminum alloy parts shall be anodized in accordance with Specification MIL-A-8625.

3.9 Identification of product. Assemblies and component parts shall be identified in accordance with Standard MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

3.10 Workmanship. The air induction system, including all parts and accessories, shall be fabricated and finished to produce a system free from defects which could affect proper functioning in service. Particular attention shall be given to the following:

(a) Freedom from blemishes, burrs, and sharp edges.

(b) Accuracy of dimensions, radii of fillets, and marking of parts and assemblies.

(c) Thoroughness of soldering, welding, brazing, painting, wiring, and riveting.

(d) Alignment of parts and tightness of assembly screws, bolts, etc.

Induction system shall be thoroughly cleaned, and free of spattered or excess solder, metal chips, and other foreign material, during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements.

4.2 Classification of tests. Inspection and testing of the air induction system shall be classified as preproduction and acceptance.

4.2.1 Preproduction. Preproduction tests shall be conducted as specified in 4.4.

4.2.2 Acceptance. Acceptance tests are those tests accomplished on air induction systems manufactured and submitted for acceptance under contract (see 4.5).

4.3 Test methods.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

4.3.1 Installation. The air induction system shall be installed in the aircraft for the applicable tests. The intercooler shall be considered a component of the air induction system for all tests conducted on this system.

4.3.2 Instrument calibration. All instruments shall be calibrated immediately before the individual tests, and at suitable intervals throughout these tests, to assure conformance of the air induction system to this specification.

4.3.3 Variations. Methods of testing other than those described herein may be used when a deviation is required by the installation and approved by the contracting officer. A report containing a brief description of the deviation(s) and data which facilitate comparisons with the normal method shall be submitted for approval by the procuring activity, prior to testing.

4.3.4 Sequence. The sequence of tests as required herein, may be varied, except that fuel metering calibration shall be conducted prior to any testing. Tests may be combined consistent with obtaining all required data for each test and accomplishing the purpose of each test.

4.4 Tests.

4.4.1 Preproduction. Preproduction tests of air induction system shall consist of all tests specified herein. The manufacturer's certification that the air induction system has previously passed these tests will be accepted instead of the actual performance of the tests; if this certification is submitted, the report of tests shall be in accordance with Standard MIL-STD-831 and shall be submitted to the contracting officer prior to production of items on contract or order.

4.4.2 Fuel metering. Fuel metering tests shall establish the absence of adverse effects of the induction and fuel system upon fuel metering. The fuel metering unit shall meet the requirements in the engine model specification or changes in design in effect at the time of conducting these tests. The unit shall have satisfactorily passed acceptance tests for production in accordance with the engine model specification and shall not have been subsequently adjusted, or tampered with, in any manner except for adjustment of idle mixture and speed, or as required. At the option of the procuring activity, a fuel metering unit that has been subjected to special metering tests shall be used for the ground and flight tests specified herein. Manifold pressure and rpm combination shall be in accordance with engine operating instructions established or approved by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

4.4.2.1 Measurements. Fuel flow measurements shall be made by either volumetric measurements, with correction for fuel density, or flowmeters which are calibrated in the aircraft immediately before the test run. Volumetric measurements may be made with small volume tanks, or by the "tank usage" method. This latter method involves filling the tank on the ground, draining at least 10 gallons to avoid loss through vent lines, and using this particular tank only for a specific time after conditions are established, then checking the quantity of fuel required to replace the fuel used. If flowmeters are used, they shall be calibrated in flight by suitable volumetric methods during sufficient runs to determine the corrections for various fuel flows, altitudes, and fuel temperature.

4.4.2.2 Procedure. Fuel metering tests shall include at least the following:

Test No.	Condition	Power	Mixtures[1]
1	Ground	Minimum Idle	Auto Rick/Auto Lean
2	Ground	40% Normal Rated	Auto Rich/Auto Lean
3	Ground	60% Normal Rated	Auto Rich/Auto Lean
4	Ground	80% Normal Rated	Auto Rich/Auto Lean[2]
5	Ground	100% Normal Rated	Auto Rich/Auto Lean[2]
6	Ground	Takeoff	Auto Rich/Auto Lean[2]
7	Flight	Minimum Cruise	Auto Rich/Auto Lean[3]
8	Flight	65% Normal Rated	Auto Rich/Auto Lean
9	Flight	Normal Rated	Auto Rich/Auto Lean[2]
10	Flight	Military	Auto Rich/Auto Lean[2]

[1]Alternate terminologies "Rich Normal" or "Full Rich Cruising Lean" may be encountered in some cases.

[2]This position to be used only when authorized by the engine operating instructions referred to in fuel metering tests.

[3]These tests shall be limited by the cylinder and coolant temperatures as specified in the engine model specification.

The above-listed flight tests shall be conducted at, or slightly below, each of the critical altitudes applicable to each power setting. In addition to conducting the fuel metering tests on the main source of engine air, engine operation shall be checked on all other sources of engine air with the mixture control in the richer of the two positions

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

available, except the "filtered air" need not be checked above 10,000 feet altitude. This procedure shall also include checks with the induction air heat valve in the following positions: full open, one-half open, full closed. Allowable fuel metering unit air temperature shall not be exceeded when using heated air source. Checks shall include:

(a) Altitude (pressure vs. density).

(b) Outside air temperature.

(c) Fuel metering air temperature.

(d) Manifold pressure.

(e) Rpm.

(f) Torquemeter pressure.

(g) Cylinder head temperature(s).

(h) Fuel pressure.

(i) Fuel flow.

(j) Airspeed.

4.4.2.3 Results. The report shall consist of a tabulation of the above-listed data for each of the runs and air source settings listed in fuel metering test procedures (see 4.4.2.2) to show conformance with fuel metering requirements of the applicable engine specification.

4.4.3 Ram recovery. The ram recovery test shall determine the ram pressure available at the air inlet of the fuel metering unit.

4.4.3.1 Instrumentation. The instrumentation shall be as required to obtain the ram air data. The air induction system shall be instrumented to determine the total pressure at the air inlet of the fuel metering unit.

4.4.3.2 Procedure. Flight tests shall be conducted at the critical altitude and military power of the aircraft, using the ram air intake. Data shall be recorded at both level flight and critical climb attitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

The following data shall be recorded, depending upon the type of system used:

(a) Nonsupercharged system.

(1) Altitude.

(2) Indicated air speed (IAS).

(3) Rpm.

(4) Manifold pressure.

(5) Aircraft free stream total pressure head.

(6) Total pressure head at the air inlet of the fuel metering unit.

(7) Torque pressure.

(8) Fuel flow.

(b) Supercharged system.

(1) Altitude.

(2) IAS.

(3) Rpm.

(4) Manifold pressure.

(5) Aircraft free stream total pressure head.

(6) Total pressure at the top deck of the fuel metering unit.

(7) Torque pressure.

(8) Fuel flow.

The pressure loss shall not exceed the values listed in 3.4.1.2.1 and 3.4.1.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

4.4.4 Induction air heat system. The induction air heat system test shall determine the air temperature rise available and the performance of the heat control.

4.4.4.1 Procedure. Tests shall be conducted at approximate sea level altitude at 65 percent normal rated power with the heat control set in the full hot position to determine conformance with 3.4.3.2 It is desired that these tests be conducted under ambient conditions as specified in 3.4.3.2; however, if this is impracticable, tests may be run at higher ambient temperatures if the data obtained are corrected for the conditions specified in 3.4.3.2. Tests shall be conducted at approximate sea level altitude: at normal rated power, 65 percent normal rated power, 50 percent normal rated power, and idle power of the engine, with the heat control set in full cold, one-quarter, one-half, three-quarter, and full hot positions, to determine conformance with 3.4.3.3.and 3.4.3.4.

4.4.4.2 Data. The following data, as well as control operating force, shall be recorded for each of the test conditions required in 4.4.2.2. procedure. Allow sufficient time for all conditions to stabilize after each change in heat control or power condition:

(a) Outside air temperature.

(b) Air temperature at fuel metering unit inlet.

(c) Total pressure at fuel metering unit inlet.

(d) Rpm.

(e) Manifold pressure.

(f) Heat control position.

(g) Cowl flap position.

(h) Time required for temperature stabilization.

(i) Force required to operate control.

(j) Torque pressure.

(k) Fuel flow.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

The airheat system and control shall conform to 3.6.1. Control forces shall not exceed the values specified in 3.4.3.1.

4.4.5 Engine induction air filter. The engine induction air filter test shall determine the effect of the filter installation on engine performance and the force necessary to operate controls.

4.4.5.1 Procedure. Flight tests shall be conducted with and without the engine charge air filter in operation. Tests shall be conducted at takeoff and at each critical altitude of the aircraft below 10,000 feet to determine effect on power available under those conditions. Tests shall be conducted to determine the critical altitudes of the aircraft using filtered air as compared to the critical altitudes using the normal "ram" air supply.

4.4.5.2 Data. The data shall be recorded at takeoff and each critical altitude with and without the filter in operation:

(a) Manifold pressure.

(b) Engine rpm.

(c) Outside air temperature.

(d) Torquemeter readings (if torquemeter is available).

(e) Altitude.

(f) Force required to actuate filter control at critical air speed.

(g) Fuel flow.

The results of the tests shall be reported. Control actuation forces shall not exceed those specified in 3.4.3.1 when the aircraft is flying at critical speed. The critical air speed is defined as that speed which shall not exceed the value stated in Specification MIL-I-8599.

4.4.6 Thermometer calibration. Test for thermometer calibration of the installation of the temperature bulb in the air induction system shall be as follows:

4.4.6.1 Instrumentation. Instrumentation shall be in accordance with figure 4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

4.4.6.2 Procedure. Static ground tests shall be conducted at 65 percent normal rated power. The thermometer shall be calibrated with the air induction heat system control in one-half, three-quarter, and full open positions. The following data shall be recorded for each condition listed:

(a) Temperature bulb reading.

(b) Thermocouple reading.

(c) Heat control position.

(d) Rpm.

(e) Manifold pressure.

(f) Outside air temperature.

4.4.7 Pressure. The pressure test shall determine the design strength of the air induction system. Instrumentation shall be as required to determine the pressures in the induction system during this test.

4.4.7.1 Procedure. The induction shall be completely sealed at the fuel metering unit for a nonsupercharged system, or at the intercooler outlet at the fuel metering unit for a supercharged system. The induction system control shall be in the unfiltered position and all outlets of the induction system positively sealed. If the nonsupercharged system is specified, the entire induction system shall be pressurized to 26 psig. The pressure shall be maintained at least 5 minutes. Any distortion or rupture shall be noted and reported. Rupture or distortion sufficient to cause unserviceability of any component shall be cause for rejection. The deformation of the intercooler, if used, shall not exceed that amount stated in 3.4.1.2.2.1.

4.4.8 High pressure leakage. High pressure leakage test shall be conducted on supercharged systems.

4.4.8.1 Procedure. The induction system shall be blocked at the entrance to the fuel metering unit and at the intercooler inlet. A pressure of 20 psig shall be applied to the system between the fuel metering unit and the intercooler inlet; the air supply shall then be shut off. The time required for the pressure to drop from 20 to 15 psig shall be recorded. The internal volume of the section of the induction system between the fuel metering unit and the intercooler inlet shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

recorded. The minimum allowable time for the pressure to drop from 20 to 15 psig shall be computed as follows:

$$T = \frac{V}{HP}$$

T = Minimum allowable time (seconds) for pressure to drop from 20 to 15 psig.

V = Internal volume (cubic feet) of the portion of the induction system tested.

HP = Military rated horsepower of the engine.

4.4.9 Intercooler effectiveness and pressure drop.

4.4.9.1 Procedure. The intercooler effectiveness determination shall be defined as follows:

$$\text{Intercooler effectiveness} = \frac{T2 - T3}{T2 - T1}$$

T1 = Temperature of cooling air entering intercooler.

T2 = Temperature of induction air entering intercooler.

T3 = Temperature of cooling air leaving intercooler.

A laboratory or flight test shall be run, using test parameters based on the assumed conditions that the engine is operated at normal rated power and the aircraft is in the maximum climb attitude at the critical altitude of the supercharger. The flow of cooling air through the intercooler will be that obtained in flight under the conditions of climb attitude at the normal rated power critical altitude as noted. The air pressure and temperature shall be assumed to vary, as shown in NASA Report No. 1235. Intercooler effectiveness as determined under the above conditions shall be as specified in 3.4.1.2.2.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

4.4.10 Intercooler vibration.

4.4.10.1 Procedure. The intercooler shall be mounted in a structure corresponding as closely as possible with respect to shape, dimensions, and material to the intercooler supporting structure to the aircraft. The supporting structure shall also be provided with the necessary stops, cushions, and pads for mounting and supporting the intercooler simulating the actual aircraft installation. If the intercooler is shock-mounted, the intercooler shall be tested in accordance with procedure XIII of Specification MIL-E-5272. If the intercooler is not shock-mounted, it shall be tested in accordance with procedure XII of Specification MIL-E-5272.

4.5 Acceptance tests. Unless otherwise specified by the procuring activity, each air induction system shall be examined to determine conformance with this specification in regard to design, construction, material, and workmanship.

5. PREPARATION FOR DELIVERY
(Not applicable.)

6. NOTES

6.1 Intended use. The air induction system is intended for use in obtaining air from outside the aircraft and conducting it to the fuel metering unit.

6.2 Supplementary data. This specification should be referenced in section 2 of the detail specification document prepared for the designated aircraft.

6.3 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type of air induction system required (see 1.2).

(c) Specified engine and aircraft for which air induction system will be installed (see 3.5.1).

(d) Instructions for delivery of data (see 6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

6.4 Definitions.

6.4.1 Critical altitude. The critical altitude for a designated engine brake horsepower (bhp) is the highest altitude at which the engine will develop the designated engine bhp in level flight without exceeding supercharger and engine model limitations as specified by the supercharger and engine model specification.

6.4.2 Maximum climb attitude. The maximum attitude is the aircraft attitude at which the maximum rate of ascent can be maintained.

6.5 Data. Data generated by this document shall not be delivered to the procuring activity unless specified on the Contract Data Requirements List (DD1423).

Custodians:

Army - AV

Preparing activity:

Army - AV

Project No. 2945-A003

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

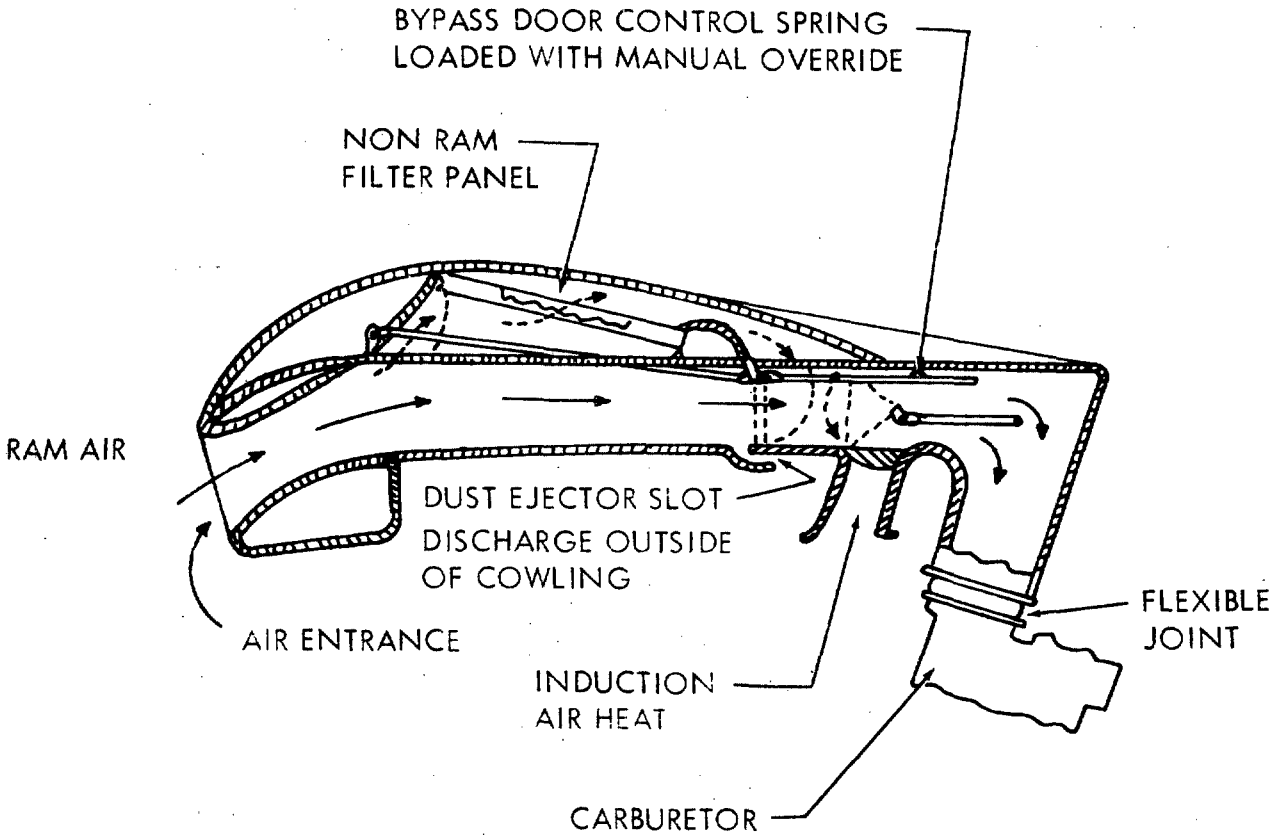


Figure 1. Typical nonsupercharger air induction system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

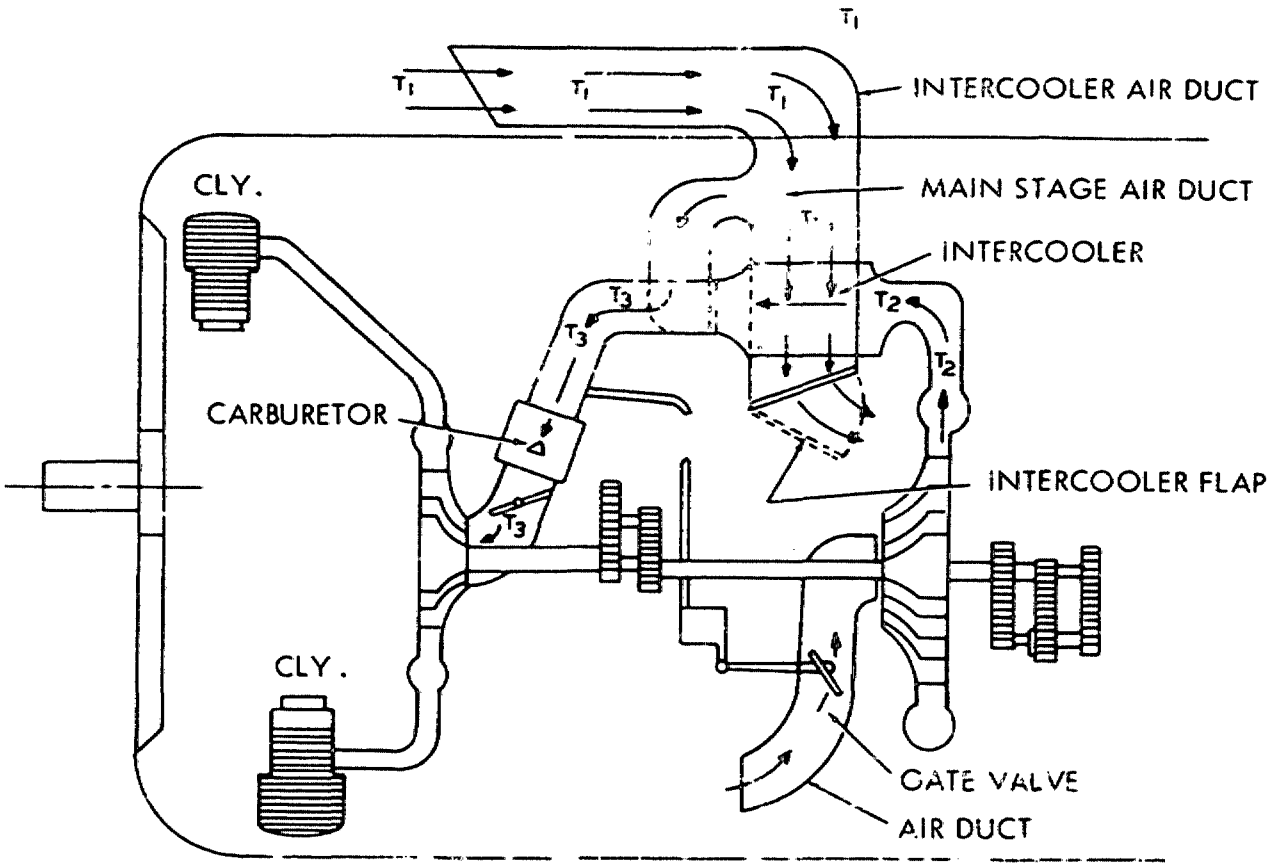


Figure 2. Typical two-stage, two speed auxiliary supercharger and intercooler air induction system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

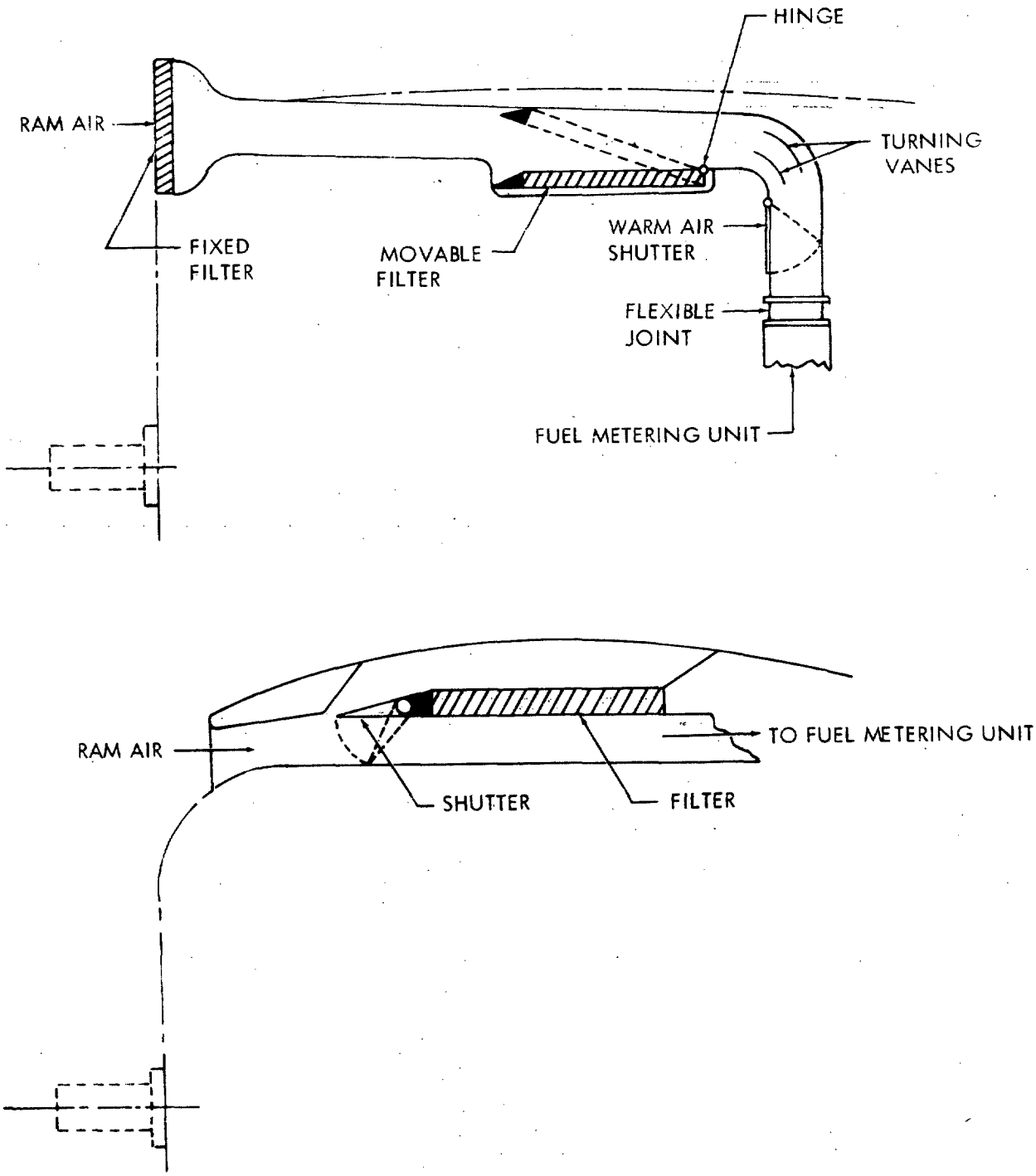


Figure 3. Typical nonsupercharged systems showing various types of ram airfilter installations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

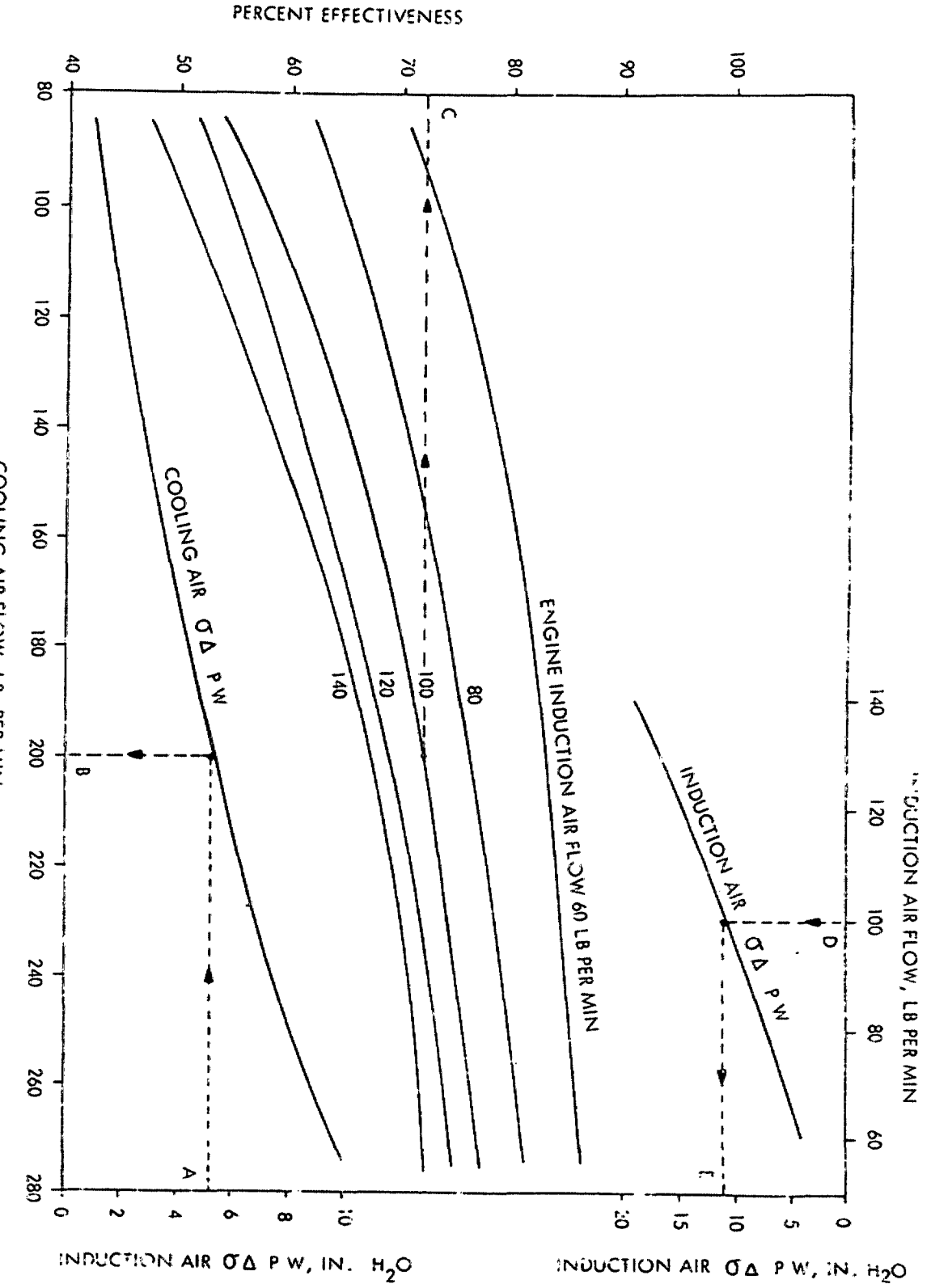


Figure 4. Typical intercooler performance curves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

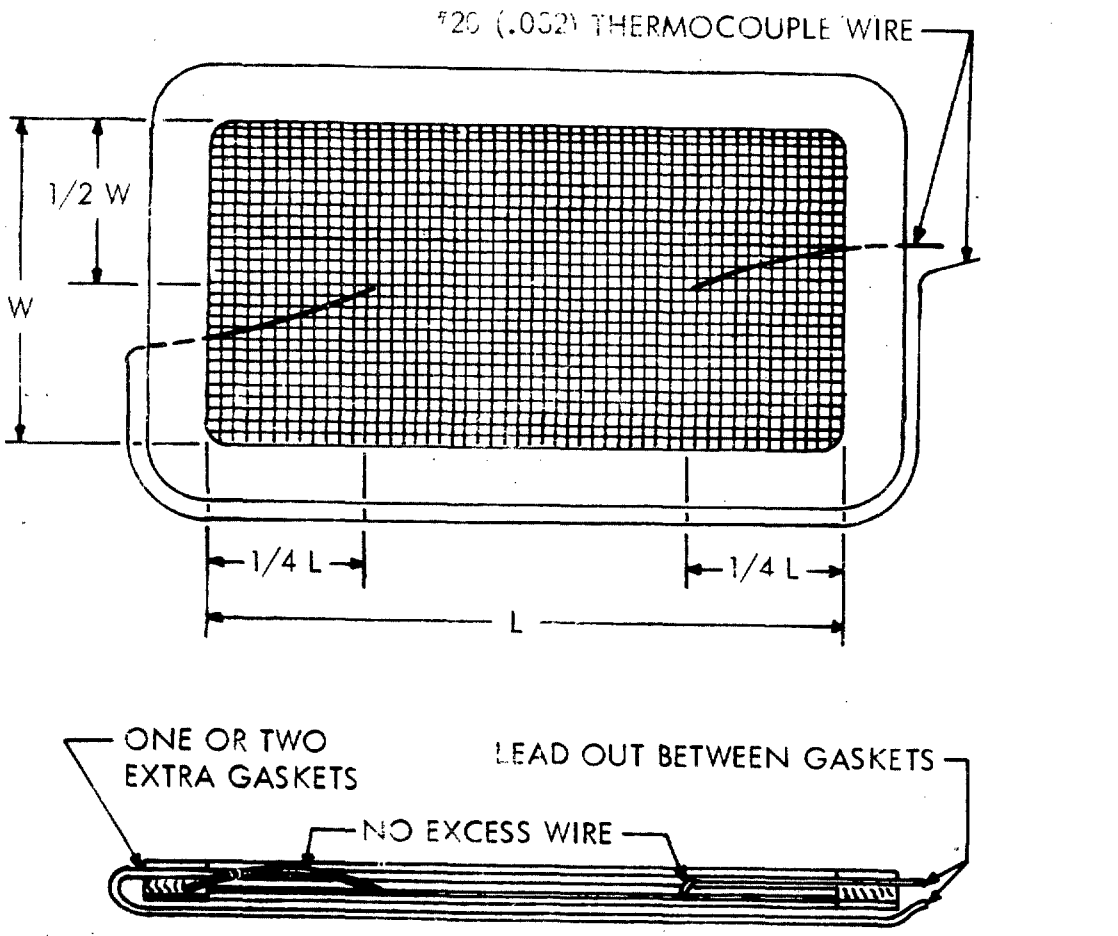


Figure 5. Installation of fuel metering unit screen thermocouple.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T08:19Z
Check the source to verify that this is the current version before use.